

11_TestPlan.md

MonitorAI v1.5 Test Plan

Version: 1.5

Status: FINAL BASELINE

1. Purpose

This document defines the validation process for MonitorAI v1.5.

The purpose is to verify:

- Clinical usability
- Technical reliability
- Dataset quality
- Upload reliability

before production deployment.

2. Testing Philosophy

MonitorAI is not tested as a camera application.

MonitorAI is tested as:

A standardized orthodontic image acquisition platform.

Success is measured by:

- Image quality
 - Metadata quality
 - Workflow completion
 - Dataset integrity
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3. Testing Categories

Required:

Unit Testing

Integration Testing

UI Testing

Device Testing

Clinical Testing

Performance Testing

Dataset Validation

4. Unit Testing

Required Services

PositionCalibrationService

OverlayGuideService

PTGService

VisibilityScoringService

AutoCaptureService

PlateauDetectionService

MetadataService

UploadService

Success Criteria

All tests pass.

Coverage target:

80%+

5. Position Calibration Testing

Verify:

Opening detection

Center alignment

Stability timer

Pass Criteria

Opening Center Error < 5%

Stable 3 Seconds

Auto Advance Triggered

6. Overlay Testing

Verify:

Correct overlay loads

Correct step mapping

Correct opacity

No distortion

Pass Criteria

11 overlays displayed correctly.

7. PTG Testing

Verify:

Dot progression

Visibility scoring

Event triggering

Pass Criteria

Dot1–Dot6 function correctly.

PTGMaxDot stored correctly.

8. Visibility Scoring Testing

Verify:

Buccal visibility scoring

Occlusal visibility scoring

Score persistence

Pass Criteria

Score range:

0–100

No invalid values.

9. Auto Capture Testing

Verify:

PTG $\geq$ Dot4

Stable 4 Seconds

Capture triggered

Pass Criteria

Capture occurs automatically.

No duplicate captures.

10. Plateau Detection Testing

Verify:

Visibility stagnation detection

Plateau timer

Plateau capture

Pass Criteria

Plateau capture triggers appropriately.

11. Voice Guidance Testing

Verify:

Prompt playback

Correct event mapping

No excessive repetition

Pass Criteria

All required prompts play correctly.

12. Haptic Testing

Verify:

Dot4

Dot5

Dot6

Capture

Upload

Pass Criteria

Expected haptic event occurs.

13. Metadata Testing

Verify every capture stores:

PTG

Visibility

Focus

Blur

Exposure

Timing

Calibration

Device Information

Pass Criteria

No missing metadata.

14. Image Metadata Testing

Verify:

ImageWidth

ImageHeight

ImageSizeBytes

ImageSHA256

Pass Criteria

Values stored for every image.

15. FileMaker API Testing

Verify:

Authentication

Session creation

Capture creation

Metadata creation

Container upload

Session completion

Pass Criteria

All API calls succeed.

16. Offline Queue Testing

Disable network.

Capture session.

Reconnect.

Pass Criteria

Images upload successfully.

No data loss.

17. Container Upload Testing

Verify:

Full image upload

Thumbnail upload

Large image handling

Pass Criteria

100% upload success.

18. Device Testing

Required Devices

iPhone 11

iPhone 12

iPhone 13

iPhone 14

iPhone 15

iPhone 15 Plus

iPhone 15 Pro

iPhone 15 Pro Max

iPhone 16 Series

Latest iPhone Available

Pass Criteria

Workflow completes successfully.

19. Orientation Testing

Verify:

Landscape Only

Rotation handling

Safe area handling

Pass Criteria

UI remains stable.

20. Clinical Testing

Real orthodontic patients.

Target:

30+

patients.

Evaluate:

Image quality

Retake frequency

Session duration

Patient understanding

Clinical usefulness

21. Workflow Completion Testing

Verify:

Calibration

11 Captures

Review

Upload

Completion

Pass Criteria

Workflow completion rate > 95%.

22. Performance Testing

Measure:

PTG refresh rate

Capture latency

Upload duration

Targets

PTG Target:

30 FPS

Minimum:

15 FPS

Auto Capture Latency:

< 500 ms

23. Dataset Validation

Verify:

Image integrity

Metadata completeness

Traceability

Duplicate detection

Pass Criteria

Every image linked to:

Patient

↓

Session

↓

Capture

↓

Metadata

24. Security Testing

Verify:

HTTPS

TLS

Token storage

Authentication

Pass Criteria

No credential exposure.

No unencrypted traffic.

25. Regression Testing

Run after every major release.

Verify:

All previous functionality remains operational.

26. Release Criteria

MonitorAI v1.5 may be released only when:

Position Calibration Pass

Overlay Pass

PTG Pass

Voice Pass

Haptics Pass

Auto Capture Pass

Plateau Detection Pass

Review Pass

Metadata Pass

Upload Pass

Offline Queue Pass

Dataset Validation Pass

27. Final Principle

Testing exists to protect:

- Clinical usefulness
- Dataset quality
- Future AI value

No feature should be released unless these goals are preserved.